Reducing Warehouse Picking Errors using Six Sigma (DMAIC)

(Inspired by real operations in companies like Amazon and DHL)

1. DEFINE

The purpose of this project is to analyze and reduce picking errors in warehouse operations. In many logistics and e-commerce companies such as Amazon and DHL, inaccurate order picking leads to increased costs, delays, and customer dissatisfaction. The problem identified in this study is a high rate of incorrect order picking.

Problem Statement:

A significant number of orders are picked incorrectly, leading to inefficiencies in warehouse operations.

Objective:

Reduce the picking error rate from 50% to below 30% and improve overall process efficiency.

2. MEASURE

A dataset of 100 orders was created to simulate warehouse picking operations.

The following parameters were measured:

1. Order accuracy (Correct / Incorrect)
2. Type of error (Wrong Item / Wrong Quantity)
3. Picking time (in minutes)

Summary of Data:

1. Total Orders: 100
2. Incorrect Orders: 50
3. Correct Orders: 50

Error Rate Calculation:

Error Rate = (Incorrect Orders / Total Orders) × 100

Error Rate = (50 / 100) × 100 = 50%

3. ANALYZE

The analysis phase focused on identifying the root causes of errors and their impact on performance.

Key Findings:

a. The error rate was calculated to be 50%, indicating a highly inefficient picking process.
b. The average picking time for correct orders was approximately 3.5 minutes.
c. The average picking time for incorrect orders was approximately 6 minutes.

This shows that incorrect orders take significantly longer to process.

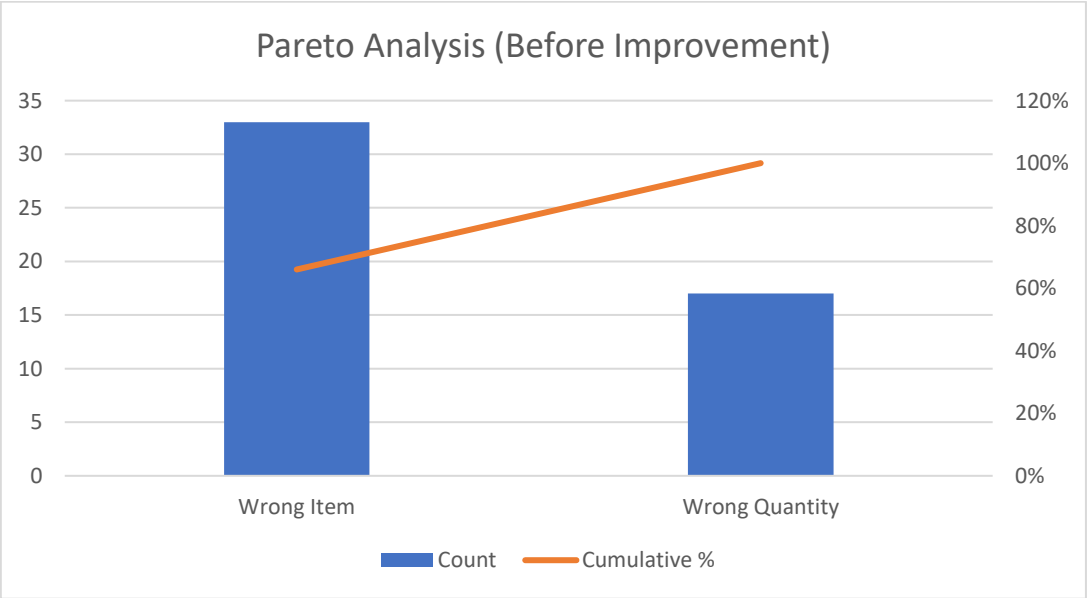
Error Type Distribution:

a. **Wrong Item**: 33 occurrences
b. **Wrong Quantity:** 17 occurrences

The majority of errors were due to wrong item selection.

Pareto Analysis (Before Improvement):

The Pareto chart below shows that "Wrong Item" errors account for the majority of picking errors.



This indicates that wrong item picking is the primary root cause and should be the focus area for improvement.

Conclusion from Analysis:

a. Wrong item picking is the primary root cause of errors.
b. Errors increase processing time, reducing overall operational efficiency.

4. IMPROVE

Based on the analysis, the following improvements were proposed:

a) Implement barcode scanning systems to ensure correct item selection
b) Improve product labeling and warehouse organization
c) Provide training to warehouse staff
d) Standardize picking procedures

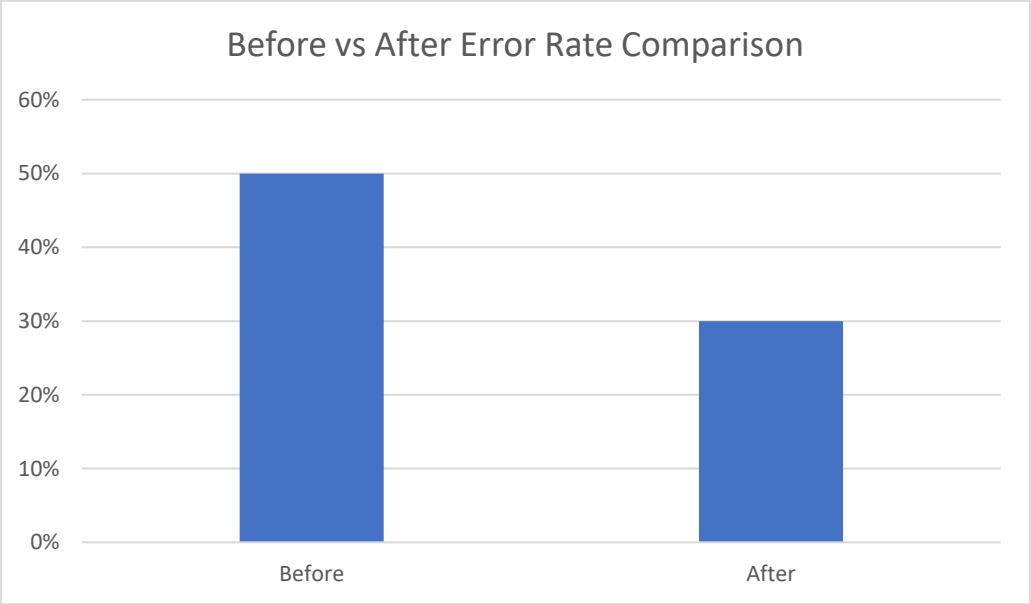
Simulated Improvement Results:

A modified dataset was created to simulate the impact of the proposed improvements by reducing wrong item errors.

a) Total errors reduced from 50 to 30
b) Error rate reduced from 50% to approximately 30%

Before vs After Comparison

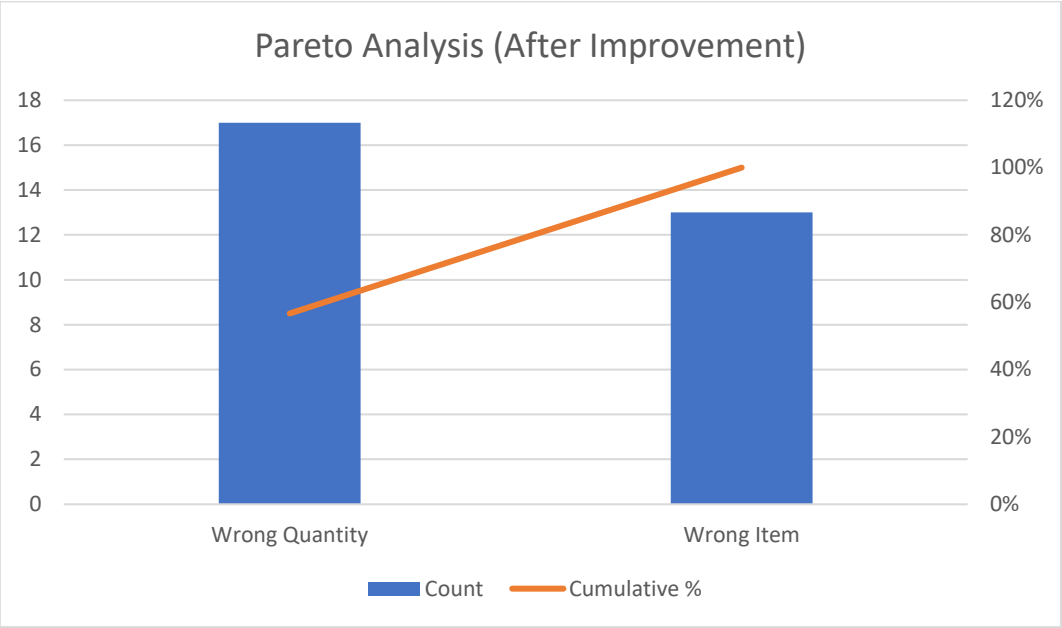
The chart below compares the error rate before and after implementing simulated improvements.



The results clearly show a reduction in error rate from 50% to approximately 30%, demonstrating the effectiveness of the proposed improvements.

Pareto Analysis (After Improvement):

The Pareto chart below illustrates the distribution of errors after implementing simulated improvements.



The contribution of wrong item errors has reduced significantly, shifting the primary issue to wrong quantity errors.

Pareto analysis confirmed that focusing on wrong item picking resulted in the most significant reduction in overall errors.

This demonstrates that addressing the primary root cause can significantly improve process efficiency.

5. CONTROL

To sustain improvements, the following control measures are recommended:

a. Monitor error rate on a weekly basis
b. Use dashboards to track key performance indicators (KPIs)
c. Implement standard operating procedures (SOPs)
d. Conduct regular audits and performance reviews

6. Conclusion

This project demonstrated how Six Sigma DMAIC methodology can be applied to identify and reduce warehouse picking errors.

By analyzing operational data, identifying root causes, and suggesting improvements, the study highlights the importance of data-driven decision-making in supply chain management.